POWER PULSE

Measurement of Current Important Formulae

Notes by Balaji Sir

1 DC ammeter with shunt

1.
$$m = \frac{I}{I_m}, \qquad R_{sh} = \frac{R_m}{m-1}$$

m = multiplying factor; I_m = full-scale meter current; R_m = meter resistance

2.
$$m = 1 + \frac{R_m}{R_{sh}}$$

3.
$$R_{in} = R_m \parallel R_{sh} = \frac{R_m R_{sh}}{R_m + R_{sh}}$$

ammeter input resistance; it should be as low as possible

2 Ammeter insertion (loading) error

4.
$$I_{read} = \frac{E}{R + R_a}, \quad I_{true} = \frac{E}{R} \quad \Rightarrow \quad \%\text{error} = -\frac{R_a}{R + R_a} \times 100$$

3 Ayrton (universal) shunt

5.
$$R_{sh,\text{range}} = \frac{R_m + R_{\text{rest}}}{m-1} \quad (\text{at a tap, the remaining shunt section is in series with the meter})$$

Ayrton shunt: several ranges from one tapped resistor; the make-before-break switch never leaves the meter unprotected

4 AC use of shunt (MI/dynamometer)

6.
$$\frac{L_m}{R_m} = \frac{L_{sh}}{R_{sh}}$$

a shunt with the same time constant as the meter divides the current equally at all frequencies

5 Rectifier ammeter

7.
$$I_{rms} = 1.11\, I_{avg} \quad (\text{full-wave, sine})$$

indicated value = form factor × average

6 High current

8.
$$I_{line} = K_n\, I_s \quad (\text{current transformer ratio})$$

a CT extends the ammeter range on AC